Macroeconomics in the Sequence Space

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April 16, 2026

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Why Sequence Space?

Sequence Space vs. State Space

Motivation

- In heterogeneous-agent (HA) models, the economy's state is the entire cross-sectional distribution — an infinite-dimensional object.
- Sequence-space methods *sidestep* this challenge: they characterize local dynamics around the steady state without ever carrying the distribution as a state variable.
- The key insight: aggregate equilibrium dynamics can be represented in sequence space as mappings from aggregate shock paths to aggregate outcomes, with Jacobians arising as the local linear approximation of these mappings.

Sequence Space vs. State Space

State Space

Definition 1

A state-space representation describes the economy recursively. At date t, a state variable x_t summarizes all payoff-relevant history, and the model evolves as

$$x_{t+1} = f(x_t, \varepsilon_{t+1}), \qquad y_t = g(x_t).$$

Given x_t, the future is independent of the past.

Sequence Space vs. State Space

Sequence Space

Definition 2

A sequence-space representation characterizes equilibrium as a mapping between entire time paths. We seek sequences $\{X_t\}_{t=0}^{\infty}$ such that equilibrium conditions hold at every date:

$$F\left(\{X_t\}_{t=0}^{\infty}, \{Z_t\}_{t=0}^{\infty}\right) = \mathbf{0},$$

where $\{X_t\}$ collects endogenous variables and $\{Z_t\}$ collects exogenous shocks or policy paths.

Sequence Space vs. State Space

Comparison

Intuition

- State space asks: *what is the current state, and how does it transition?*
- Sequence space asks: *given a shock path, what equilibrium outcome path clears all markets?*

Takeaway

When the distribution is the intractable state variable, sequence space replaces a high-dimensional state-space problem with a representation in terms of equilibrium mappings over aggregate paths, making analysis and computation much more tractable.

Local Solution

Example: The Krusell-Smith Model

Setup

- The economy is populated by a mass 1 of heterogeneous households,
- Each household maximizes its expected lifetime utility:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right],$$

- Standard CRRA utility,
- There exist n_e idiosyncratic states w/ transition matrix $P(e, e')$ and stationary distribution π,
- Agents supply an exogenous number of hours n, and earn wage income $w_t e n$,
- Agents can only trade in capital k, which pays a rental rate r_t net of depreciation.

Example: The Krusell-Smith Model

Household Problem

The value function of an agent entering period w/ state (e, k_-) is given by:

$$
\begin{aligned}
V_t(e, k_-) &= \max_{c,\, k} u(c) + \beta \sum_{e'} V_{t+1}(e', k)\, P(e, e') \\
\text{s.t.} \qquad c + k &= (1 + r_t)k_- + w_t e n \\
k &\geq 0
\end{aligned}
$$

- Denote by $c_t^*(e, k_-)$ and $k_t^*(e, k_-)$ the policy functions that solve the Bellman equation.
- Also denote by $D_t(e, K_-)$ the measure of households in state e that own capital in a set K_- at the start of date t.

Example: The Krusell-Smith Model

Household Problem

The distribution D_t has law of motion

$$D_{t+1}(e', K) = D_t(e, k_t^{*-1}(e, K))P(e, e').$$

where $k_t^{*-1}(e, K)$ denotes the inverse of $k_t^*(e, K)$.

- We assume that prior to $t = 0$, the economy is in a steady state.
- We suppose that agents start in this stationary distribution at date 0, so that $D_0 = D_{ss}$.

Example: The Krusell-Smith Model

Aggregate Capital

- For any t, the policy $k_t^*(e, k_-)$ is a function of the future path $\{r_s, w_s\}_{s=t}^{\infty}$.
- Given $D_0 = D_{ss}$, the distribution $D_t(e, K)$ at any t is a function of the entire path $\{r_s, w_s\}_{s=0}^{\infty}$.
- The aggregate household capital holdings are characterized by a capital function $\mathcal{K}_t(\{r_s, w_s\}_{s=0}^{\infty})$.

$$\mathcal{K}_t\left(\{r_s, w_s\}_{s=0}^{\infty}\right) = \sum_e \int_{k_-} k_t^*(e, k_-) D_t(e, dk_-)$$

Key Insight

The ability to reduce interactions between heterogeneous agents to functions such as $\mathcal{K}_t$, which map aggregate sequences into aggregate sequences, is key to the sequence-space Jacobian method.

Example: The Krusell-Smith Model

Production

- Firm adopts a Cobb-Douglas production function:

$$Y_t = Z_t K_t^{\alpha} N_t^{1-\alpha}$$

- The firm's first-order conditions:

$$r_t = \alpha Z_t K_{t-1}^{\alpha-1} N_t^{1-\alpha} - \delta$$
$$w_t = (1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}$$

Key Insight

The paths of prices $\{r_t, w_t\}_{t=0}^{\infty}$ are now functions of the exogenous paths $\{Z_t, N_t\}_{t=0}^{\infty}$ and the endogenous path $\{K_t\}_{t=0}^{\infty}$.

Example: The Krusell-Smith Model

Capital Market-Clearing Condition

Combining together, we can express the capital market-clearing condition at each point in time as a function H_t:

$$H_t(\boldsymbol{K}, \boldsymbol{Z}) \equiv \mathcal{K}_t \left(\left\{ \underbrace{\alpha Z_s \left(\frac{K_{s-1}}{\sum \pi(e) e n} \right)^{\alpha-1} - \delta}_{r_s}, \underbrace{(1-\alpha) Z_s \left(\frac{K_{s-1}}{\sum \pi(e) e n} \right)^{\alpha}}_{w_s} \right\}_{s=0}^{\infty} \right) - K_t = 0$$

where $\boldsymbol{K}$ is the vector of capital holdings $\{K_t\}_{t=0}^{\infty}$ and $\boldsymbol{Z}$ is the vector of exogenous shocks $\{Z_t\}_{t=0}^{\infty}$.

Equilibrium Capital Path

Given initial capital K_{-1} and the exogenous path for productivity $\boldsymbol{Z}$, we have equilibrium path of capital $\{K_t\}_{t=0}^{\infty}$ by solving the nonlinear equation $H_t(\boldsymbol{K}, \boldsymbol{Z}) = 0$.

Example: The Krusell-Smith Model

IRF

By *implicit function theorem*, given an exo. shock $\mathrm{d}\boldsymbol{Z} = (\mathrm{d}Z_0, \mathrm{d}Z_1 \ldots)'$ at $t = 0$, the linear impulse response of capital to this shock is given by:

$$d\boldsymbol{K} = -H_K^{-1} H_Z \, d\boldsymbol{Z}$$

where H_K and H_Z denote the Jacobians of H with respect to $\boldsymbol{K}$ and $\boldsymbol{Z}$, evaluated at the steady state.

- Given $\mathrm{d}\boldsymbol{K}$, the impulse responses of other variables—for example, $\{Y_t, r_t, w_t\}_{t=0}^{\infty}$—follow immediately.
- In practice, $H_t(\boldsymbol{K}, \boldsymbol{Z}) = 0$ is solved up to a sufficiently large horizon T such that $\boldsymbol{K}$ and $\boldsymbol{Z}$ have approximately returned to their steady states by time T.

Example: The Krusell-Smith Model

Constructing the Jacobian

- We use the *chain rule* to relate the Jacobians H_K and H_Z to the derivatives of the $\mathcal{K}$.
- For example, differentiating $H_t(\boldsymbol{K}, \boldsymbol{Z}) = 0$ with respect to K_s
- The (t, s) entry of H_K is:

$$[H_K]_{t,s} = \frac{\partial \mathcal{K}_t}{\partial r_{s+1}} \frac{\partial r_{s+1}}{\partial K_s} + \frac{\partial \mathcal{K}_t}{\partial w_{s+1}} \frac{\partial w_{s+1}}{\partial K_s} - \mathbf{1}_{\{s=t\}}$$

- A similar expression applies to H_Z.
- In addition, the derivatives such as $\frac{\partial r_{s+1}}{\partial K_s}$ at (K^{ss}, Z^{ss}) can all be computed analytically.
- For example:

$$\frac{\partial r_{s+1}}{\partial K_s} = \alpha(\alpha - 1) Z^{ss} (K^{ss})^{\alpha-2} (N^{ss})^{1-\alpha}$$

Example: The Krusell-Smith Model

Constructing the Jacobian

Recall that:

$$d\boldsymbol{K} = -H_K^{-1} H_Z \, d\boldsymbol{Z},$$

To get $H_K^{-1} H_Z$, the only thing left is to solve the Jacobians of the $\mathcal{K}$ function with respect to its two inputs r and w.

$$[H_K]_{t,s} = \frac{\partial \mathcal{K}_t}{\partial r_{s+1}} \frac{\partial r_{s+1}}{\partial K_s} + \frac{\partial \mathcal{K}_t}{\partial w_{s+1}} \frac{\partial w_{s+1}}{\partial K_s} - \mathbf{1}_{\{s=t\}}$$

Let

$$\mathcal{J}_{t,s}^{\mathcal{K},r} = \frac{\partial \mathcal{K}_t}{\partial r_s}, \quad \mathcal{J}_{t,s}^{\mathcal{K},w} = \frac{\partial \mathcal{K}_t}{\partial w_s}$$

be the Jacobians up to some truncation horizon T.

Example: The Krusell-Smith Model

Constructing the Jacobian

Sufficient statistics

The Jacobians $\mathcal{J}^{\mathcal{K},r}$ and $\mathcal{J}^{\mathcal{K},w}$ are sufficient statistics for the household in sequence space equilibrium:

- Shocks propagate through the dynamics of the household capital distribution, but *we do not need those details if we have the Jacobians.*
- Column s of $\mathcal{J}^{\mathcal{K},r}$ is the impulse response of capital to a news shock dr_s; the Jacobian gives these for all $s = 0, \ldots, T-1$.
- These are rich objects, but *there is a fast new algorithm to compute them.*

Fake News Algorithm

General Model Representation

We represent a heterogeneous-agent problem as a mapping from a time path of aggregate inputs $\{\boldsymbol{X}_t\}$ to a time path of aggregate outputs $\{\boldsymbol{Y}_t\}$.

Assume:

- $\boldsymbol{X}_t \in \mathbb{R}^{n_x}$: aggregate inputs,
- $\boldsymbol{Y}_t \in \mathbb{R}^{n_y}$: aggregate outputs,
- the distribution is discretized on n_g grid points,
- $\boldsymbol{D}_t \in \mathbb{R}^{n_g}$: cross-sectional distribution of agents at time t.

Aggregate outputs are averages of individual outcomes against the distribution:

$$\boldsymbol{Y}_t = \boldsymbol{y}_t' \boldsymbol{D}_t,$$

where $\boldsymbol{y}_t$ is an $n_g \times n_y$ matrix of individual outcomes (for example, consumption, savings, labor supply, or other individual-level objects).

Fake News Algorithm

Three Core Equations

We assume there exist three functions v, Λ, and y such that, for a given initial distribution $\boldsymbol{D}_0$, the aggregate sequence solves

$$\boldsymbol{v}_t = v(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t),$$

$$\boldsymbol{D}_{t+1} = \Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)\, \boldsymbol{D}_t,$$

$$\boldsymbol{Y}_t = y(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)' \boldsymbol{D}_t.$$

Interpretation:

- $\boldsymbol{v}_t$: backward-looking object, typically the value function;
- $\Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)$: transition matrix governing the law of motion of the distribution;
- $y(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)$: individual outcomes used to aggregate into $\boldsymbol{Y}_t$.

Fake News Algorithm

Steady State and Transition Paths

For a constant aggregate input $\boldsymbol{X}_{ss}$, the steady state is a fixed point

$$(\boldsymbol{Y}_{ss}, \boldsymbol{v}_{ss}, \boldsymbol{D}_{ss})$$

of the system.

For convenience, define

$$\Lambda_{ss} \equiv \Lambda(\boldsymbol{v}_{ss}, \boldsymbol{X}_{ss}), \qquad y_{ss} \equiv y(\boldsymbol{v}_{ss}, \boldsymbol{X}_{ss}).$$

We consider transition paths of length T that end at the steady state:

$$\boldsymbol{X}_{T-1} = \boldsymbol{X}_{ss}, \qquad \boldsymbol{v}_T = \boldsymbol{v}_{ss}.$$

The initial distribution $\boldsymbol{D}_0$ is given and often set equal to $\boldsymbol{D}_{ss}$.

Hence, this framework is designed to study transitory shocks around a steady state.

Fake News Algorithm

Economic Meaning of the System

Step 1: Backward

Given the future object $\boldsymbol{v}_{t+1}$ and current aggregates $\boldsymbol{X}_t$, solve for the current object $\boldsymbol{v}_t$.

Step 2: Forward

Use $\Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)$ to update the cross-sectional distribution:

$$\boldsymbol{D}_{t+1} = \Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)\boldsymbol{D}_t.$$

Step 3: Aggregate

Aggregate individual outcomes across agents:

$$\boldsymbol{Y}_t = y(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)'\boldsymbol{D}_t.$$

Fake News Algorithm

Three-Step Structure of the HA Model

Fake News Algorithm

Three-Step Structure of the HA Model

Step 1: Backward

$$\boldsymbol{v}_t = v(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)$$

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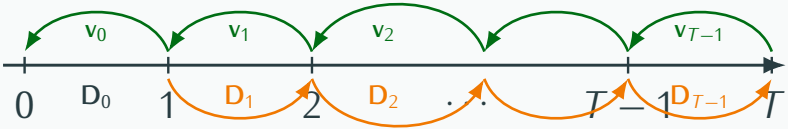

Step 2: Forward

$$\boldsymbol{D}_{t+1} = \Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)\boldsymbol{D}_t$$

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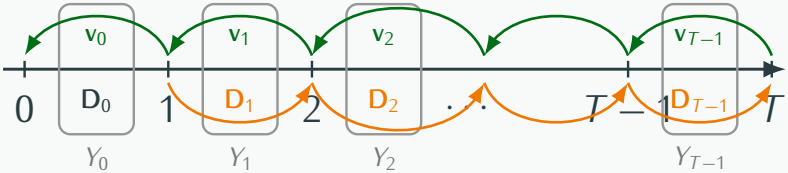
Three-Step Structure of the HA Model

Step 1: Backward

$$\boldsymbol{v}_t = v(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)$$



Step 2: Forward

$$\boldsymbol{D}_{t+1} = \Lambda(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)\boldsymbol{D}_t$$

Step 3: Aggregate

$$\boldsymbol{Y}_t = y(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t)'\boldsymbol{D}_t$$

Fake News Algorithm

Calculating the Jacobian: direct vs. new algorithm

- The direct algorithm requires a separate backward and forward iteration for each $s = 0, \ldots, T-1$; this is costly.
- SSJ method uses a single backward and forward iteration to get all T columns.
- *This improves speed by a factor of roughly T with no loss in accuracy.*
- The key is to exploit time symmetries in the linearized system.

The linearized system:

$$
\begin{aligned}
d\boldsymbol{Y}_t &= d\boldsymbol{y}_t' \boldsymbol{D}_{ss} + \boldsymbol{y}_{ss}' \, d\boldsymbol{D}_t \\
d\boldsymbol{D}_{t+1} &= d\Lambda_t \boldsymbol{D}_{ss} + \Lambda_{ss} d\boldsymbol{D}_t \\
\boldsymbol{D}_0 &= d\boldsymbol{D}_{ss}
\end{aligned}
$$

Since everything is linear, we can look at terms from backward and forward iterations separately, then combine them.

Fake News Algorithm

Terms from backward iteration

Holding $\boldsymbol{D}_t = \boldsymbol{D}_{ss}$ fixed, what is the effect of $d\boldsymbol{X}_s$?

$$d\boldsymbol{Y}_t = d\boldsymbol{y}_t^0 \boldsymbol{D}_{ss} + \cancel{\boldsymbol{y}'_{ss}\, d\boldsymbol{D}_t}$$
$$d\boldsymbol{D}_{t+1} = d\Lambda_t^0 \boldsymbol{D}_{ss} + \cancel{\Lambda_{ss}\, d\boldsymbol{D}_t}$$

- $d\boldsymbol{y}_t^0$ and $d\Lambda_t^0$ are purely forward-looking from the Bellman equation.
- They depend only on the time until the shock, $s - t$.

Define:

$$\mathcal{Y}_{s-t} \equiv \frac{d\boldsymbol{y}_t^0 \boldsymbol{D}_{ss}}{d\boldsymbol{X}_s} \qquad \mathcal{D}_{s-t} \equiv \frac{d\Lambda_t^0 \boldsymbol{D}_{ss}}{d\boldsymbol{X}_s}$$

Use a single backward iteration with a shock $d\boldsymbol{X}_s$ at $s = T - 1$ to obtain $d\boldsymbol{y}_t^0$ and $d\Lambda_t^0$ for all $t = T - 1, \ldots, 0$.

Fake News Algorithm

Terms from forward iteration

Holding $\boldsymbol{y}_t = \boldsymbol{y}_{ss}$ and $\Lambda_t = \Lambda_{ss}$ fixed, what is the effect of $d\boldsymbol{D}_t$?

$$d\boldsymbol{Y}_t = \cancel{d\boldsymbol{y}_t^0 \boldsymbol{D}_{ss}} + \boldsymbol{y}'_{ss}\, d\boldsymbol{D}_t$$
$$d\boldsymbol{D}_{t+1} = \cancel{d\Lambda_t^0 \boldsymbol{D}_{ss}} + \Lambda_{ss} d\boldsymbol{D}_t$$

Since everything's linear, some linear functional (row vector) maps $d\boldsymbol{D}_t$ to $d\boldsymbol{Y}_t$: what is it?

- Can write for $s \leq t$ (zero for $s > t$):

$$d\boldsymbol{Y}_t = \underbrace{\boldsymbol{y}'_{ss}(\Lambda'_{ss})^{t-s}}_{\equiv \mathcal{P}'_{t-s}} d\boldsymbol{D}_s$$

- These $\mathcal{P}_u$ can be calculated recursively as:

$$\mathcal{P}_u = \Lambda'_{ss}\mathcal{P}_{u-1}$$

- Thus, a single "transpose" forward iteration gives all $\mathcal{P}_u$.

Fake News Algorithm

Putting it together: the Jacobian

For shock $d\boldsymbol{X}_s$, we want:

$$d\boldsymbol{Y}_t = d\boldsymbol{y}_t'\boldsymbol{D}_{ss} + \boldsymbol{y}_{ss}'\, d\boldsymbol{D}_t$$

Change in policy, holding distribution constant:

$$d\boldsymbol{y}_t'\boldsymbol{D}_{ss} = \mathcal{Y}_{s-t} d\boldsymbol{X}_s$$

Change in distribution, holding policy constant:

$$\boldsymbol{y}_{ss}'\, d\boldsymbol{D}_t = (\mathcal{P}_0'\mathcal{D}_{s-t+1} + \ldots + \mathcal{P}_{t-1}'\mathcal{D}_s) d\boldsymbol{X}_s$$

sum of shocks $\mathcal{D}$ to distribution from anticipating $d\boldsymbol{X}_s$ before t, propagated forward to effect on $d\boldsymbol{Y}_t$ with $\mathcal{P}$

The full Jacobian is then:

$$\mathcal{J}_{t,s} = \mathcal{Y}_{s-t} + (\mathcal{P}_0'\mathcal{D}_{s-t+1} + \ldots + \mathcal{P}_{t-1}'\mathcal{D}_s)$$

Fake News Algorithm

Fake news matrix

Define the fake news matrix as:

$$\mathcal{F}_{t,s} \equiv \begin{cases} \mathcal{Y}_s & \text{if } t = 0, \\ \mathcal{P}'_{t-1}\mathcal{D}_s & \text{if } t > 0. \end{cases}$$

- Interpretation:
  - $t = 0$: agents get news of shock at date $s \geq 0$
  - $t \geq 1$: news of unrealized shock disappears ("fake news")
- Build Jacobian (response to news shocks) recursively from fake news matrix (response to fake news shocks) as:

$$\mathcal{J}_{t,s} \equiv \begin{cases} \mathcal{F}_{t,s} & \text{if } t = 0 \text{ or } s = 0, \\ \mathcal{F}_{t,s} + \mathcal{J}_{t-1,s-1} & \text{otherwise} \end{cases}$$

Fake News Algorithm

Algorithm

To obtain Jacobians $\mathcal{J}^{o,i}$ for many inputs dX^i and outputs dY^o:

1. For each input i, perform backward iteration with T steps to get all $\dagger_u^{o,i}$ and $\mathcal{D}_u^i$.
2. For each output o, perform transpose forward iteration with $T-1$ steps to get all $\mathcal{P}_u^o$.
3. For each pair (o, i), construct the fake news matrix $\mathcal{F}^{o,i}$ from $\dagger^{o,i}$ in the first row and the product $(\mathcal{P}^o)'\mathcal{D}^i$ for other rows.
4. For each pair (o, i), recursively compute $\mathcal{J}^{o,i}$ from $\mathcal{F}^{o,i}$.

Step 1 is almost always the bottleneck.

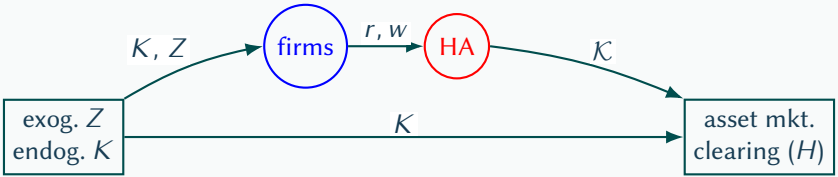
DAG

Krusell-Smith model as a directed acyclic graph (DAG)

Recall that we wrote equilibrium in KS model as system

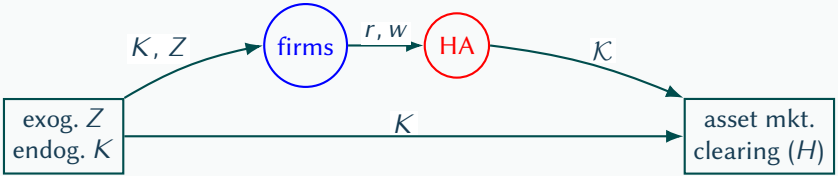
$$H_t(\boldsymbol{K}, \boldsymbol{Z}) \equiv \mathcal{K}_t \left(\left\{ \underbrace{\alpha Z_s \left(\frac{K_{s-1}}{\sum \pi(e) e n} \right)^{\alpha-1} - \delta}_{r_s}, \underbrace{(1-\alpha) Z_s \left(\frac{K_{s-1}}{\sum \pi(e) e n} \right)^{\alpha}}_{w_s} \right\}_{s=0}^{\infty} \right) - K_t = 0$$

This corresponds to a simple graph:



DAG

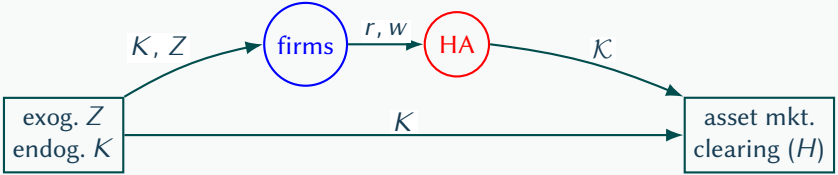
DAGs combining simple and HA blocks



- Each node ("block") takes sequences as inputs and outputs.
- Inputs of later blocks are outputs of earlier blocks.
- The number of endogenous variables ("unknowns") equals the number of equations in H ("targets"); solve to get equilibrium.
- Two kinds of blocks in SHADE models:
  - Simple blocks: Analytical equations directly in aggregates, e.g. $r_t = \alpha Z_t K_{t-1}^{\alpha-1} - \delta$; Jacobians are sparse and easy to obtain.
  - HA blocks: As described previously.

DAG

DAGs combining simple and HA blocks



- Define $\mathbf{J}^{o,i}$ to be the total derivative of o along the DAG with respect to the exogenous or endogenous variable i (distinct from the block Jacobian $\mathcal{J}^{o,i}$).
- Here, for example, $\mathbf{J}^{r,K}$ equals $\mathcal{J}^{r,K}$, etc.: the direct effect is the only effect.
- Then we need the chain rule. For instance:

$$\mathbf{J}^{\mathcal{K},K} = \mathcal{J}^{\mathcal{K},r}\mathbf{J}^{r,K} + \mathcal{J}^{\mathcal{K},w}\mathbf{J}^{w,K}$$

- Finally, we obtain $\mathbf{H}_K$ and $\mathbf{H}_Z$; then $\mathbf{G}^{K,Z} \equiv -\mathbf{H}_K^{-1}\mathbf{H}_Z$ gives the general equilibrium map from Z to K, and all $\mathbf{G}^{o,Z}$ are determined from it.

DAG

General case: solving with forward accumulation

- Initialize $\mathbf{J}^{i,i}$ as the identity for all exogenous or unknown i.
- Evaluate the chain rule following a topological sort:

$$\mathbf{J}^{o,i} = \sum_{m \in I_b} \mathcal{J}^{o,m} \mathbf{J}^{m,i}$$

  on blocks b to get $\mathbf{H}_U = \mathbf{J}^{H,U}$ and $\mathbf{H}_Z = \mathbf{J}^{H,Z}$.
- This is called "forward accumulation" in the algorithmic differentiation literature.
- Compute the general equilibrium mapping: $\mathbf{G}^{U,Z} = -\mathbf{H}_U^{-1}\mathbf{H}_Z$
- Then do forward accumulation again:

$$\mathbf{G}^{o,Z} = \sum_{m \in I_b} \mathcal{J}^{o,m} \mathbf{G}^{m,Z}$$

  to get general equilibrium mappings $\mathbf{G}^{o,Z}$ from shocks dZ to all variables o of interest.
- This gives impulse responses to every shock simultaneously.

Determinacy and Large-Scale Solutions in the Sequence Space

From truncated Jacobians to quasi-Toeplitz operators

Main result

For stationary models, sequence-space Jacobians admit the decomposition

$$J = T(j) + E, \qquad j(z) = \sum_{k=-\infty}^{\infty} j_k z^k,$$

where $T(j)$ is Toeplitz and E is a compact correction.

- Far from date 0, the Jacobian is approximately translation invariant:

$$J_{t,s} \approx j_{t-s}.$$

- E captures missing anticipation: shocks near $t = 0$ are not fully anticipated in the past.
- This applies to stationary heterogeneous-agent blocks & stable linear expectational systems.

Determinacy and Large-Scale Solutions in the Sequence Space

Determinacy in sequence space

To solve equilibrium, we need

$$Jx = y.$$

For block quasi-Toeplitz J, define the matrix-valued symbol $j(z)$ and study $\det j(z)$ on the unit circle.

Winding-number test

$$\operatorname{wind}(\det j) = \operatorname{nonex}(J) - \operatorname{indet}(J).$$

Generically:

$$\begin{cases} \operatorname{wind}(\det j) = 0 & \Rightarrow \text{unique solution}, \\ \operatorname{wind}(\det j) < 0 & \Rightarrow \text{indeterminacy}, \\ \operatorname{wind}(\det j) > 0 & \Rightarrow \text{nonexistence}. \end{cases}$$

- Extends Onatski's Toeplitz test to the quasi-Toeplitz case.
- Equivalent to Blanchard–Kahn in standard linear systems, but works in sequence space.

Determinacy and Large-Scale Solutions in the Sequence Space

Large-scale solution

The idea is to exploit the Toeplitz core instead of directly inverting a huge truncated matrix.

Preconditioned iterative solution

Use an approximate inverse

$$P \approx T(j^{-1}) + \widetilde{E}$$

as a preconditioner, and solve the truncated system iteratively (e.g. GMRES).

- Toeplitz part is cheap to apply/store:

$$T_T(j)x \quad \text{via FFT in } O(T \log T).$$

- Compact correction is compressed in low rank:

$$J_{T,T} \approx T_T(j) + U_{T,r} V'_{T,r}, \qquad r \ll T.$$

- This reduces truncation error and avoids direct $nT \times nT$ inversion.

Determinacy and Large-Scale Solutions in the Sequence Space

Takeaway (Auclert et al., 2025a)

What this adds to SSJ

The paper extends SSJ in two directions:

- theory: existence / uniqueness / determinacy;
- computation: very large sequence-space systems can be solved quickly.

2nd SSJ

From certainty equivalence to certainty correspondence

Known result

At first order, which is the usual certainty equivalence

$$\text{perfect-foresight sequence space} \iff \text{linear stochastic MA}(\infty),$$

Compare:

$$X_t(\{\sigma Z_s\}_{s\geq 0}, D_{ss}) \qquad \text{and} \qquad \widetilde{X}_t = \mathcal{X}_t(\sigma, \epsilon_t, \epsilon_{t-1}, \ldots; \{Z_s\}).$$

Certainty correspondence

$$\text{2nd-order perfect-foresight responses} \iff \text{coefficients of a quadratic MA}(\infty) \text{ for } \widetilde{X}_t.$$

So SSJ is still enough beyond certainty equivalence.

2nd SSJ

What corresponds to what?

Expand the stochastic solution in σ:

$$\widetilde{X}_t = X + \sum_{j\geq 0} x_j \epsilon_{t-j} + \frac{1}{2}\left[\mathcal{X}_{\sigma\sigma}(0)\sigma^2 + \sum_{j,k\geq 0} \mathcal{X}_{\epsilon_{-j}\epsilon_{-k}}(0)\, \epsilon_{t-j}\epsilon_{t-k}\right] + O(\sigma^3).$$

By symmetry in σ,

$$\mathcal{X}_{\sigma\epsilon_{-j}}(0) = 0.$$

Interpretation

- $\{x_j\}$ are the familiar 1st-order perfect-foresight IRFs.
- The quadratic coefficients

  $$\mathcal{X}_{\epsilon_{-j}\epsilon_{-k}}(0)$$

  are identified by 2nd-order perfect-foresight responses.

2nd SSJ

Why does this matter?

Rewrite the quadratic MA as

$$\widetilde{X}_t \simeq X + \sum_{j\geq 0} x_j \epsilon_{t-j} + \frac{1}{2}\mathcal{X}_{\sigma\sigma}(0)\sigma^2 + \frac{1}{2}\sum_{j\geq 0} \mathcal{X}_{\epsilon_{-j}\epsilon_{-j}}(0)\epsilon_{t-j}^2 + \sum_{j<k} \mathcal{X}_{\epsilon_{-j}\epsilon_{-k}}(0)\epsilon_{t-j}\epsilon_{t-k}.$$

- Constant term: Anticipated risk.
- ϵ^2 terms: Size dependence.
- $\epsilon_{t-j}\epsilon_{t-k}$ terms: State dependence.

2nd SSJ

Takeaway (Auclert et al., 2025b)

What this adds to SSJ

The paper extends SSJ in two directions:

- Extends SSJ from linear IRFs to nonlinear, state-dependent dynamics in sequence space;
- A practical framework for studying risk, nonlinear propagation, and welfare in rich heterogeneous-agent models.

SSJ in Continuous Time

What changes relative to discrete-time SSJ?

Core comparison

Discrete-time SSJ / FNA:

$$\text{discretize first} \implies \text{differentiate numerically.}$$

Continuous-time SSJ:

$$\text{differentiate in continuous time} \implies \text{discretize later.}$$

- The paper keeps the same sequence-space logic as Auclert et al.: household response → distribution → fake news → Jacobian.
- The main change is only in the first step: instead of numerically perturbing a nonlinear Euler equation, the household response is characterized analytically in continuous time.

SSJ in Continuous Time

Takeaway (Bilal and Goyal, 2025)

- Faster: Jacobians and impulse responses are about 3 times faster to compute.
- Where does the gain come from? From the policy-function step only: linear PDEs replace nonlinear backward solves plus numerical differentiation.
- Why continuous time helps: with a binding borrowing constraint, the first-order condition still holds with equality, so the perturbation is cleaner.

Broader contribution

The paper shows that SSJ / FNA is not tied to discrete time. It extends sequence-space methods naturally to continuous-time HA/HANK models, especially environments with rich heterogeneity and occasionally binding constraints.

Applications of the extended version of SSJ

- Fiscal and monetary policy (Auclert et al., 2025c, Auclert et al., 2024b)
- Het households + het firms (Winberry et al., 2025)
- Behavioural macro (Angeletos et al., 2025, Auclert et al., 2020)
- Life-cycle model (Bardoczy and Velasquez Giraldo, 2025, Bardóczy and Velásquez-Giraldo, 2024)
- Dynamic spatial model (Donald et al., 2025, Fan et al., 2023)
- Production network (Schaab and Tan, 2023)
- Het banks (Bellifemine et al., 2022)
- Optimal policy (Auclert et al., 2024a, McKay and Wolf, 2022)
- Macro labor (Guerreiro et al., 2024, Kekre, 2023)

Global Solution

Repeated transition method

From local SSJ to global sequence space

What is the extension?
Standard SSJ is mainly a local method:

$$\text{1st-order} \;+\; \text{perfect foresight} \;+\; \text{IRFs around a reference point.}$$

RTM extends sequence-space methods to

$$\text{global} \;+\; \text{nonlinear} \;+\; \text{aggregate uncertainty.}$$

- It does not assume perfect foresight.
- It does not require a parametric law of motion for the aggregate distribution.

Repeated transition method

Why "repeated transitions"?

To solve backward at time t, agents need a conditional expectation of continuation values:

$$E_t V_{t+1} = \Pi_{S_t,G} V(\cdot; \Phi_{t+1}, G) + \Pi_{S_t,B} V(\cdot; \Phi_{t+1}, B).$$

RTM's trick

If the simulated path is long enough, a similar aggregate state reappears. Find a period

$$\tilde{t} + 1 = \arg\min_{\tau} \Phi_{\tau}^{(n)} - \Phi_{t+1}^{(n)}$$

with the counterfactual shock realization. Then use the realized value function at $\tilde{t} + 1$ to fill in the missing continuation value.

- Intuition: history repeats itself along an ergodic equilibrium path.
- So expectations are built from repeated transitions, not from an estimated law of motion.

Repeated transition method

Takeaway (Lee, 2025)

- More general than SSJ: can study global nonlinear dynamics, not only local IRFs.
- Avoids law-of-motion misspecification: no need to summarize the distribution with a guessed forecasting rule.

Takeaway

RTM keeps the sequence-space perspective, but turns it into a global stochastic solution method: better suited for nonlinear state dependence, occasionally binding constraints, and models with non-trivial market clearing.

DL in sequence space

From local SSJ to global sequence space

What is the extension?
Standard SSJ is mainly a local method:

$$\text{aggregate state} \xrightarrow{\text{Jacobian}} \text{local IRFs.}$$

This paper instead builds a global approximation to equilibrium in sequence space.

In place of a high-dimensional aggregate state such as (z_t, μ_t), use a truncated shock history

$$z_t^T = (z_{t-T}, z_{t-T+1}, \ldots, z_t),$$

- In ergodic economies, the truncation error vanishes as T grows.
- So the aggregate state is approximated by a sequence of past aggregate shocks, not by moments or histograms of the distribution.

DL in sequence space

How deep learning enters

Parameterize equilibrium objects—policies or prices—with a neural network:

$$\mathcal{N}_\rho(X_t^{\text{seq},T}) \approx \text{equilibrium object at } t.$$

Train the network to satisfy equilibrium conditions along simulated ergodic paths:

$$\rho^* = \arg\min_\rho \frac{1}{N} \sum_{t=1}^{N} \left(\text{equilibrium residual}_t(\rho)\right)^2.$$

- The network input is the truncated history of aggregate shocks.
- The training data come from simulated paths of the economy.

Compared with SSJ

Instead of computing derivatives around a reference point, the method directly learns nonlinear equilibrium mappings over the ergodic set.

DL in sequence space

Takeaway (Azinovic-Yang and Žemlička, 2025)

- Handles richer models globally: large shocks, aggregate uncertainty, and nonlinearities.
- Scales to high-dimensional environments: OLG, heterogeneous households & firms, and life-cycle models with discrete retirement choice.
- Shape-preserving architectures: predicted policies can be forced to be monotone (or concave), which makes EGM usable inside the algorithm.

Takeaway

The paper extends sequence-space methods from fast local perturbation to global nonlinear equilibrium computation, using deep learning to make long shock histories computationally tractable.

RL in sequence space

Agents learn prices (Yang et al., 2025)

The key step is to replace the high-dimensional distribution state by low-dimensional prices:

$$p_t = P^*(g_t, z_t), \qquad \pi_t = \pi(s_t, z_t, p_t).$$

- Households care directly about prices, not the full distribution.
- The algorithm therefore works with prices as state variables.
- This sidesteps the Master equation, where the distribution would enter the Bellman equation.

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